

Measuring Marxian Categories in the U.S. Economy: Profitability, Distribution, and Capital Composition

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Abstract

Empirical work on the rate of profit depends on how theoretical categories are mapped into national accounts. Conventional accounts treat every marketed activity as production, whereas the Marxian framework distinguishes activities that produce new value from those that circulate or appropriate it. Building on the framework of Shaikh and Tonak (1994), we classify every industry in the U.S. input-output accounts into production, semi-production, business, financial, and non-production sectors, justify the classification of borderline industries, and construct the Marxian net product, surplus value, profit, the rate of surplus value, the organic composition of capital, and four measures of the rate of profit for 1997–2023. Three findings stand out. First, the distinction between production and other activities is quantitatively large and growing: the share of the net product recorded in production sectors fell from 59.5 to 55 percent, while the share recorded in business sectors that realize output rose from 36 to 41 percent. The Marxian net product is about 70 percent of GDP but roughly one and a half times as volatile. Second, the level of the rate of profit depends heavily on the treatment of unproductive labor and capital, ranging from 9 to 42 percent across measures, although all measures move together. Third, an exact decomposition shows that the organic composition of capital rose by about 0.9 percent a year and the rate of surplus value by about 0.7 percent a year, so that the Marxian rate of profit shows a weak downward tendency that is not statistically significant, with sharp falls in each recession. The share of surplus value absorbed by the wages of unproductive labor declined from about two-thirds to one-half, which is why conventional profit rates rose while the Marxian rate of profit did not.

Keywords: rate of profit, productive and unproductive labor, Marxian net product, rate of surplus value, input-output accounts

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1 Introduction

The rate of profit occupies a central place in classical and Marxian political economy. It links production, technical change, competition, and accumulation. In Marx’s account, the pursuit of surplus value drives capitalists to mechanize production, raising the organic composition of capital. Because only living labor creates surplus value, this tends to lower the general rate of profit, a tendency that “at a certain point comes into antagonistic conflict” with the development of the productive forces and “must be overcome constantly through crises” (Marx, 1998, p. 257). When profitability falls far enough, capitalists scale back investment and hiring, lower wage income depresses consumption, and falling demand discourages investment further, setting off a cumulative downturn (Sweezy, 1942; Foley, 1986; Heinrich, 2012; Basu, 2018). Whether the rate of profit actually shows such a tendency is therefore an empirical question with direct implications for the theory of crisis.

A large empirical literature has accordingly measured the U.S. rate of profit over the postwar period. Some studies find a sustained decline (Weisskopf, 1979; Wolff, 1986; Basu and Manolakas, 2013; Tsoulfidis and Paitaridis, 2019), while others find an initial rise followed by a decline (Moseley, 1990).

These disagreements are partly about measurement. Marxian categories such as surplus value, variable capital, and the organic composition of capital are defined in terms of productive activity, but official national accounts are built on a different production boundary. In the national accounts, any activity that is marketed, or that the government provides, contributes to output. In the Marxian framework, only activities that transform use values into new use values produce value; activities that transfer existing products from sellers to buyers, or that maintain and reproduce the social order, are financed out of the value produced elsewhere (Shaikh and Tonak, 1994; Wolff, 1987; Moseley, 1992). Measures of the rate of profit computed directly from official aggregates therefore mix categories that the theory keeps distinct, and conclusions drawn from them may depend as much on accounting conventions as on economic dynamics.

This paper takes the measurement problem as its starting point. Following Shaikh and Tonak (1994), we reconstruct Marxian categories from the U.S. national income and product accounts and the input-output (IO) accounts of the Bureau of Economic Analysis (BEA). Our contribution is to apply the framework at the level of individual IO industries and to make the classification of each industry explicit. Rather than adopting a broad sector-level classification, we assign every industry in the IO accounts to one of five groups: production, semi-production, business, financial, and non-production sectors. We discuss the industries whose classification is contested and show how sensitive the results are to alternative choices. We then compute the Marxian net product (MNP), surplus value, profit, the rate of surplus value, the organic composition of capital (OCC), and four alternative measures of the rate of profit for 1997–2023, a period that includes three recessions.

The exercise yields three findings. First, the distinction between production and other activities is quantitatively consequential. Only 54–60 percent of the reconstructed MNP is recorded in production sectors in the IO accounts, while 35–42 percent is recorded in business sectors such as

wholesale and retail trade, which realize rather than produce value, and the business share has risen since the late 1990s. The Marxian net product is about 70 percent of GDP but is roughly one and a half times as volatile: in 2009 it fell by 6.5 percent, compared with 2.0 percent for nominal GDP, because the activities that GDP counts but that are financed out of value produced elsewhere continued to grow. Second, the level of the rate of profit depends heavily on the treatment of unproductive labor and capital. Our four measures range from about 9 to 42 percent, although they move together over time and fall in every recession. Third, an exact decomposition of the Marxian rate of profit into the rate of surplus value and the organic composition of capital shows that both rose over the period, the organic composition slightly faster, so that the Marxian rate of profit shows a weak downward tendency that is not statistically significant. The share of surplus value absorbed by the wages of unproductive labor fell from about two-thirds to one-half, so conventional profit rates rose while the Marxian rate of profit did not. Our estimated rate of surplus value, 1.2–1.6, is well below the estimates of Tsoulfidis and Paitaridis (2019), a difference we trace to the definition of variable capital.

The rest of the paper is organized as follows. Section 2 reviews how the Marxian and national-accounts production boundaries differ. Section 3 presents the classification of activities and industries, discusses borderline cases, and defines the measured categories. Section 4 describes the data. Section 5 presents the results, compares the Marxian net product with GDP, analyzes long-run movements, examines their sensitivity to measurement choices, and compares them with previous estimates. Section 6 concludes.

2 Production Boundaries in National and Marxian Accounts

National accounts measure gross domestic product in three equivalent ways: as the sum of value added across industries, as the sum of incomes generated in production, and as the sum of expenditures on final goods and services. Each method rests on the same production boundary, under which essentially all marketed activities, together with government services, count as production (Shaikh and Tonak, 1994). Wholesale and retail trade, finance, legal services, and general government therefore all contribute value added.

The Marxian framework draws the boundary differently. Production is the activity through which use values are transformed into new use values, and only labor employed in such activity creates value and surplus value. Activities that transfer commodities and money between owners, such as trade and finance, and activities that maintain the social order, such as general government, law, and the military, use up part of the social product rather than adding to it (Wolff, 1987; Moseley, 1992; Shaikh and Tonak, 1994). Their incomes are paid out of the value created in production.

Two examples illustrate how the boundaries diverge. First, the national accounts impute a rental value to owner-occupied housing and include it in output, even though no market transaction takes place. Second, the compensation of general government employees is counted as value added, although government revenue derives from taxation and borrowing, that is, from transfers of value

produced elsewhere. From a Marxian perspective, both conventions overstate the value produced in the economy. More generally, because official aggregates count the value added of circulation and social-maintenance activities as output, they misstate both the net product and its division between wages and surplus value.

Shaikh and Tonak (1994) develop a systematic method for translating national accounts into Marxian categories, and subsequent work has applied it to the U.S. economy (Mohun, 2005; Paitaridis and Tsoulfidis, 2012; Tsoulfidis and Paitaridis, 2019); see Basu (2015) for a review. Mohun (2005), for example, measures productive labor using the Standard Industrial Classification without separately assessing whether each industry’s activity is productive. The adoption of the North American Industry Classification System (NAICS) has since made it possible to separate many service activities that the older classification grouped together (Tsoulfidis and Paitaridis, 2019). Our approach classifies each industry in the IO accounts directly, which makes the mapping between theoretical categories and data transparent at the industry level.

3 Framework

3.1 Classification of activities

Following Shaikh and Tonak (1994), we distinguish four types of activity in social reproduction. *Production* uses up use values to create new use values. *Distribution* uses up use values to transfer other use values from their current holders to those who intend to use them. *Social maintenance and reproduction* uses up use values in the administration and reproduction of the social order, through government, the legal system, and the military. *Personal consumption* is the direct consumption of use values by individuals. All four types of activity may employ labor, but only production yields output in the Marxian sense. Labor employed in production is therefore productive labor, while labor employed in distribution and social maintenance is unproductive labor, compensated out of income generated in production.

3.2 Classification of industries

Shaikh and Tonak (1994) group industries into two broad sectors. Production and trade together form the *primary flows* sector, whose gross output constitutes the total value generated in the economy. Industries in the primary flows sector earn their revenue by producing and selling commodities. The revenues of all other industries, such as ground rent, interest and finance charges, fees, royalties, insurance charges, and taxes, are derived from the primary flows. They are either a recirculation of money flows generated in the primary flows, a circulation of socially validated claims on part of those flows, or both. These industries form the *secondary flows* sector, and their “output” does not add to the new value created in the economy (Shaikh and Tonak, 1994; Basu, 2015).

We refine this two-sector distinction by assigning every industry in the BEA IO accounts to one of five groups, based on the classification of activities above. *Production sectors* are those engaged in the generation of surplus value in the pursuit of profit, excluding industries engaged in distribution,

circulation, and the maintenance of existing relations of production. *Semi-production sectors* also carry out production activity, but part of their output is purchased by the government or financed through government expenditure. Only the part of their output that is realized through the market is counted as production.

The remaining three groups do not produce value. *Business sectors* engage primarily in the realization of value, that is, in changing its form, as when commodities are converted into money. Wholesale trade, retail trade, and rental and leasing belong to this group. *Financial sectors* appropriate part of the surplus value produced by productive capital through their control over money capital. Finally, *non-production sectors* derive their income from value created in production and subsequently redistributed. They include general government as well as private activities that maintain existing relations of production, such as legal services.

This finer classification separates activities that the primary/secondary distinction groups together: it distinguishes the realization of value in trade from its appropriation in finance, and it allows for production that is only partly carried out for the market. The full assignment of industries is reported in Table 8 in the Appendix.

3.3 Borderline industries

The classification of most industries follows directly from the definitions above. Agriculture, mining, utilities, construction, and manufacturing transform material inputs into new use values, while trade, finance, and general government transfer, appropriate, or administer value produced elsewhere. Several industries, however, are contested in the literature, and because our contribution rests on the classification of individual industries, we state the reasoning behind each of these choices.

Transportation and warehousing. Marx treats the transport industry as productive, since changing the location of a use value is part of completing it for consumption, and the costs of transport add value to the commodity (Marx, 1997). Storage that preserves use values is likewise a continuation of production rather than a pure cost of circulation. We therefore classify all transportation industries and warehousing and storage as production sectors.

Market services. Educational services, health care, social assistance, the arts, accommodation, and food services produce use values that are sold as commodities, even though these use values are consumed as they are produced. Following Shaikh and Tonak (1994) and Tsoulfidis and Paitaridis (2019), we classify them as production. The same reasoning applies to information industries, such as publishing, broadcasting, and data processing, and to computer systems design and other professional, scientific, and technical services, which sell services that enter into production or final consumption. An important feature of the IO accounts makes this treatment consistent with our definition of semi-production. Public schools and public hospitals are recorded under state and local general government rather than under educational services and hospitals. The latter industries therefore consist mainly of providers selling their output on the market, and the publicly provided part of education and health care is already included in general government, which we classify as non-production.

Other services. The industry “other services, except government” combines market services, such as repair and personal services, with the activities of religious, civic, and membership organizations, which are financed largely by contributions rather than sales. The IO accounts do not separate these activities, so we treat the industry as semi-production and assume that half of its activity is productive ($\theta = 0.5$ below). Section 5.5 shows that the results are insensitive to this choice.

Real estate and rental. Owner-occupied and rental housing is classified as non-production. The national accounts impute a rent to owner-occupied housing that involves no market transaction, and the income from residential rental is a claim on value produced elsewhere. Other real estate, which consists mainly of brokerage and the leasing of commercial property, and rental and leasing services, which include the lessors of intangible assets, realize and circulate value through the transfer of use rights and are therefore classified as business sectors.

Management, administrative support, and legal services. The management of companies and enterprises, administrative and support services, and legal services perform supervisory, administrative, and protective functions. These functions are necessary for capitalist enterprise, but they do not transform use values. Their costs correspond to what business accounting calls overhead, which Tsoulfidis and Paitaridis (2019) argue is paid out of surplus value. We classify these industries as non-production.

Government enterprises and accounting adjustments. Federal and state and local government enterprises, such as the postal service and public utilities and transit, sell their output on the market and are classified as production. The items “scrap, used and secondhand goods” and “noncomparable imports and rest-of-the-world adjustment” are accounting adjustments rather than activities and do not contribute to the Marxian net product.

3.4 Mapping sectors into the Marxian net product

We measure output and value added at producers’ prices, which include taxes on products net of subsidies. This choice follows Shaikh and Tonak (1994), for whom indirect taxes are part of the value produced and are appropriated by the state out of surplus value. Valuing output and value added at basic prices, which exclude these taxes, would lower the levels of the Marxian net product and the rate of surplus value but not their movements over time. Because the IO accounts record transactions at producers’ prices, the value realized through trade appears as the output of the business sectors rather than as part of the value added of the producing industries. The value added of production sectors therefore understates the net product they generate, and part of that net product is recorded in other sectors. Our measurement strategy is to identify these transfers and add them back. Table 1 summarizes the mapping.

Let NVA_i denote the net value added (value added minus depreciation) of industry i , X_i its total output, G_i government expenditure on its output, and z_{ji} the payments from industry i to financial industry j . Denote by $\mathcal{P}$, $\mathcal{S}$, $\mathcal{B}$, and $\mathcal{F}$ the sets of production, semi-production, business,

Table 1: Activities, sectors, and their contribution to the Marxian net product

Activity	Sector	Contribution to MNP
Production	Production	Net value added
	Semi-production	Productive share of net value added
Distribution	Business	Total industry output
Social maintenance	Financial	Payments from production sectors and productive share of payments from semi-production sectors
	Non-production	None
Personal consumption	–	None

and financial industries. The productive share of a semi-production industry is

$$\theta_i = \frac{X_i - G_i}{X_i}, \quad i \in \mathcal{S}. \quad (1)$$

The Marxian net product is then

$$MNP = \sum_{i \in \mathcal{P}} NVA_i + \sum_{i \in \mathcal{S}} \theta_i NVA_i + \sum_{i \in \mathcal{B}} X_i + \sum_{j \in \mathcal{F}} \left(\sum_{i \in \mathcal{P}} z_{ji} + \sum_{i \in \mathcal{S}} \theta_i z_{ji} \right). \quad (2)$$

The first two terms are the net product recorded in production and semi-production sectors. The third term is the value realized through the business sectors: since the IO accounts record their output at producers' prices, this output originates in the production and semi-production sectors. The fourth term adds the part of the net product paid to the financial sectors by producing industries. Payments from business sectors to financial sectors are excluded, because the full output of business sectors is already included and adding them would double count. Non-production sectors consume rather than produce value, so their recorded output is excluded.

The payments z_{ji} are measured as the purchases of financial commodities by each producing industry, that is, as the entries of the Use table in the rows of the four financial commodities and the columns of the production and semi-production industries. These entries record what producing industries pay for financial services, which is the flow that equation (2) requires: the part of the net product generated in production that is appropriated by finance. Because the Use table records commodities rather than the industries that produce them, we also trace these purchases to financial *industries* using the market shares implied by the Make table. The industry-based payments exceed the commodity-based ones by 5 to 13 percent, but because payments to finance account for only 2.5 to 3.3 percent of the Marxian net product, the adjustment has a negligible effect on the results (Section 5.5).

The IO accounts record no government expenditure for the industries that would otherwise be classified as semi-production, so θ_i cannot be computed from the accounts. As discussed in Section 3.3, the semi-production group consists of a single industry, other services except government, for which we set $\theta = 0.5$.

3.5 Measured categories

Table 2 defines the categories we construct. Productive compensation is the compensation of employees in production sectors plus the productive share of compensation in semi-production sectors. It is the empirical counterpart of variable capital. Surplus value is the Marxian net product minus productive compensation. Profit is the Marxian net product minus total compensation of employees.

The difference between surplus value and profit is conceptually important. Profit treats all wages as a cost of production, as in conventional accounts. Surplus value treats only the wages of productive workers as variable capital, so that the wages of unproductive workers are counted as a use of surplus value rather than as a deduction from it, as in Shaikh and Tonak (1994). Surplus value therefore always exceeds profit, and the gap between them equals the compensation of unproductive labor.

Table 2: Definitions and data sources

Symbol	Category	Definition	Source
MNP	Marxian net product	Equation (2)	BEA IO Use and Make tables
CE	Compensation of employees	Total compensation of employees	BEA IO Use tables
CE^p	Productive compensation	Compensation in production sectors plus productive share in semi-production sectors	BEA IO Use tables
S	Surplus value	$MNP - CE^p$	Derived
Π	Profit	$MNP - CE$	Derived
e	Rate of surplus value	S/CE^p	Derived
K	Capital stock	Current-cost net stock of private nonresidential fixed assets	BEA Fixed Assets Table 1.1
K^p	Productive capital stock	Current-cost net stock of production industries plus the productive share of semi-production industries	BEA Fixed Assets Tables 3.1ESI and 7.1
OCC	Organic composition of capital	K^p/CE^p	Derived

We measure productive capital directly, as the current-cost net stock of fixed assets of the production industries plus the productive share of the semi-production industry, using the BEA estimates of net stocks by industry. The net stock of government enterprises, which the Fixed Assets accounts report only in total, is divided between federal and state and local enterprises in proportion to their value added. We prefer this direct measure to the common alternative of allocating the total private nonresidential capital stock in proportion to each sector's share of depreciation, which understates productive capital for two reasons. First, the depreciation of real estate includes the depreciation of residential structures, which are excluded from the nonresidential capital stock, so the depreciation shares attribute too much of that stock to the business sector. Second, depreciation rates differ widely across assets, so that industries whose capital consists mainly of equipment and software receive a larger share of capital than their net stocks warrant. Measured directly,

productive capital is 34 to 45 percent larger than the depreciation-based estimate. The level of the Marxian rate of profit is correspondingly lower, by about 16 percentage points, although the two measures move closely together (Section 5.5).

We compute four measures of the rate of profit that combine the two numerators with the two capital stocks:

$$R_1 = \frac{\Pi}{K}, \quad R_2 = \frac{S}{K}, \quad R_3 = \frac{\Pi}{K^p}, \quad R_4 = \frac{S}{K^p}. \quad (3)$$

R_1 is closest to conventional measures of the profit rate. R_4 is closest to the Marxian concept, relating surplus value to the capital advanced in production.

The Marxian measure can be decomposed exactly into the two ratios on which the law of the tendency of the rate of profit to fall turns. Dividing the numerator and denominator of R_4 by productive compensation gives

$$R_4 = \frac{S/CE^p}{K^p/CE^p} = \frac{e}{OCC}, \quad (4)$$

so that the Marxian rate of profit rises with the rate of surplus value and falls with the organic composition of capital. The other measures are related to R_4 through $R_2 = R_4 K^p/K$ and $\Pi = S - (CE - CE^p)$: the gap between surplus value and profit is the compensation of unproductive labor.

4 Data

We use the BEA summary-level input-output Use and Make tables before redefinitions, at producers' prices, for 1997–2023. These tables report value added, its components, industry output, and interindustry transactions for 71 industries on a classification that is consistent over the whole period, so the classification in Table 8 applies unchanged to every year. Total industry output is taken from the Make table. The capital stock, depreciation, and net stocks by industry come from the BEA Fixed Assets accounts (Tables 1.1, 3.1ESI, 3.4ESI, 7.1, and 7.3). All values are in current dollars.

5 Results

5.1 The Marxian net product and its sectoral distribution

Table 3 reports the constructed categories. The Marxian net product rises from \$6.0 trillion in 1997 to \$20.1 trillion in 2023, with declines in 2009 and 2020.

Table 4 and Figure 1 show where the Marxian net product is recorded in the IO accounts. Production sectors account for 57–60 percent of the Marxian net product until the Great Recession and for 54–56 percent since 2019, while the share recorded in business sectors rises from 36 to 41 percent. Financial sectors account for 2.5–3.3 percent and the semi-production sector for 1.3–1.9 percent. In other words, roughly two-fifths of the net product generated in production appears in

Table 3: Marxian categories and rates of profit, 1997–2023

Year	MNP (\$bn)	e	OCC	R_1	R_2	R_3	R_4	$(CE - CE_p)/S$
1997	5,994	1.35	3.23	13.2%	35.6%	15.6%	41.8%	0.63
1998	6,287	1.29	3.14	11.8%	34.6%	14.0%	41.0%	0.66
1999	6,724	1.28	3.09	12.1%	34.8%	14.4%	41.5%	0.65
2000	7,143	1.24	3.05	11.1%	34.1%	13.3%	40.8%	0.67
2001	7,175	1.21	3.14	9.3%	32.2%	11.1%	38.5%	0.71
2002	7,436	1.29	3.26	10.2%	33.2%	12.2%	39.6%	0.69
2003	7,868	1.36	3.29	11.6%	34.8%	13.7%	41.4%	0.67
2004	8,571	1.44	3.39	13.0%	35.7%	15.5%	42.4%	0.64
2005	9,186	1.49	3.52	13.7%	35.6%	16.2%	42.3%	0.62
2006	9,733	1.49	3.60	13.4%	34.8%	15.9%	41.3%	0.61
2007	10,118	1.46	3.63	12.6%	34.0%	14.9%	40.2%	0.63
2008	10,198	1.42	3.77	11.3%	31.8%	13.4%	37.6%	0.64
2009	9,538	1.36	3.86	9.7%	30.2%	11.3%	35.2%	0.68
2010	10,165	1.49	3.96	11.9%	32.5%	13.8%	37.5%	0.63
2011	10,666	1.50	3.99	12.4%	32.7%	14.3%	37.6%	0.62
2012	11,188	1.49	3.93	12.9%	33.2%	14.8%	38.0%	0.61
2013	11,781	1.54	3.97	14.0%	33.9%	16.0%	38.8%	0.59
2014	12,352	1.53	3.95	14.0%	33.9%	16.0%	38.7%	0.59
2015	12,851	1.50	3.82	14.0%	34.2%	16.0%	39.2%	0.59
2016	13,216	1.49	3.81	14.0%	34.1%	16.0%	39.0%	0.59
2017	13,859	1.49	3.78	14.2%	34.4%	16.3%	39.4%	0.59
2018	14,689	1.49	3.74	14.8%	34.8%	16.9%	39.9%	0.58
2019	15,277	1.46	3.71	14.5%	34.3%	16.6%	39.4%	0.58
2020	14,698	1.37	3.81	11.5%	31.4%	13.1%	35.9%	0.64
2021	17,178	1.54	3.85	15.3%	34.6%	17.7%	39.9%	0.56
2022	19,165	1.61	3.95	17.0%	35.1%	19.7%	40.8%	0.52
2023	20,127	1.58	3.85	17.0%	35.4%	19.7%	41.0%	0.52

Notes: BEA summary Use and Make tables before redefinitions at producers' prices (1997–2023) and BEA Fixed Assets Tables 1.1, 3.4ESI, and 7.3. Payments to financial sectors are purchases of financial commodities by production and semi-production industries; $\theta = 0.5$.

the official accounts as the output of sectors that realize and circulate value, and this share has grown.

Table 4: Share of the Marxian net product recorded in each sector (percent)

Year	Production	Semi-production	Business	Financial	MNP (\$bn)
1997	59.5	1.8	35.9	2.8	5,994
2000	58.8	1.8	36.2	3.2	7,143
2007	58.2	1.5	37.2	3.2	10,118
2009	59.8	1.6	35.3	3.3	9,538
2017	56.1	1.4	39.8	2.7	13,859
2019	56.1	1.4	39.7	2.8	15,277
2020	54.8	1.4	40.9	2.9	14,698
2021	54.2	1.3	42.0	2.5	17,178
2022	54.3	1.3	41.8	2.6	19,165
2023	55.2	1.3	40.9	2.6	20,127

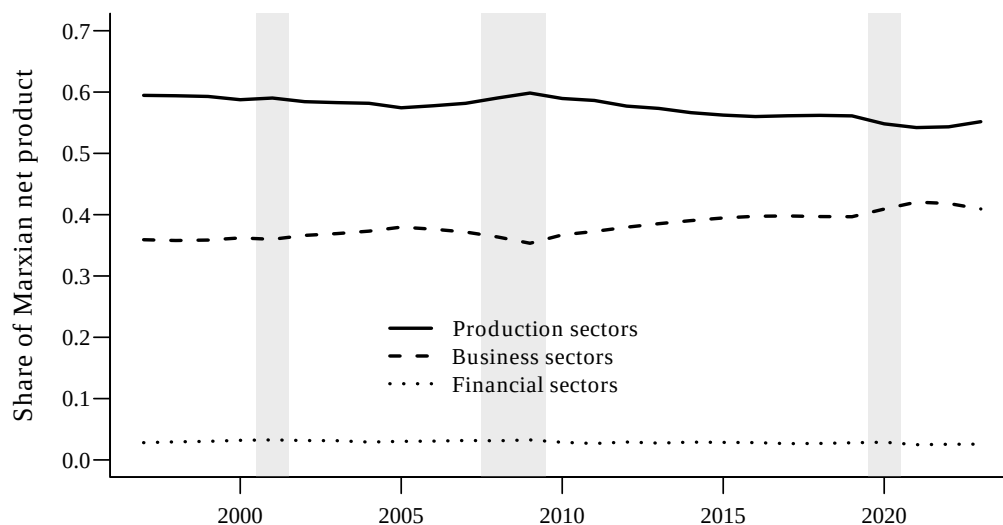


Figure 1: Share of the Marxian net product recorded in each sector, 1997–2023. Shaded areas are recession years.

The division of compensation follows a similar pattern. Productive compensation accounts for 52–55 percent of total compensation of employees in every year, so that nearly half of all wages and salaries are paid to workers in distribution, finance, and social maintenance. Production industries hold 84–88 percent of the private nonresidential capital stock, and the remainder is employed in business, financial, and non-production activities.

5.2 Four measures of the rate of profit

Figure 2 and Table 3 report the four profit rates. Their levels differ substantially. The conventional-type measure R_1 lies between 9 and 17 percent, while R_4 , which relates surplus value to productive

capital, lies between 35 and 42 percent. Against the total capital stock, using surplus value rather than profit in the numerator raises the rate by 18 to 23 percentage points, and using productive rather than total capital in the denominator raises it by a further 4 to 7 points. The level of the rate of profit is therefore highly sensitive to the treatment of unproductive labor and capital.

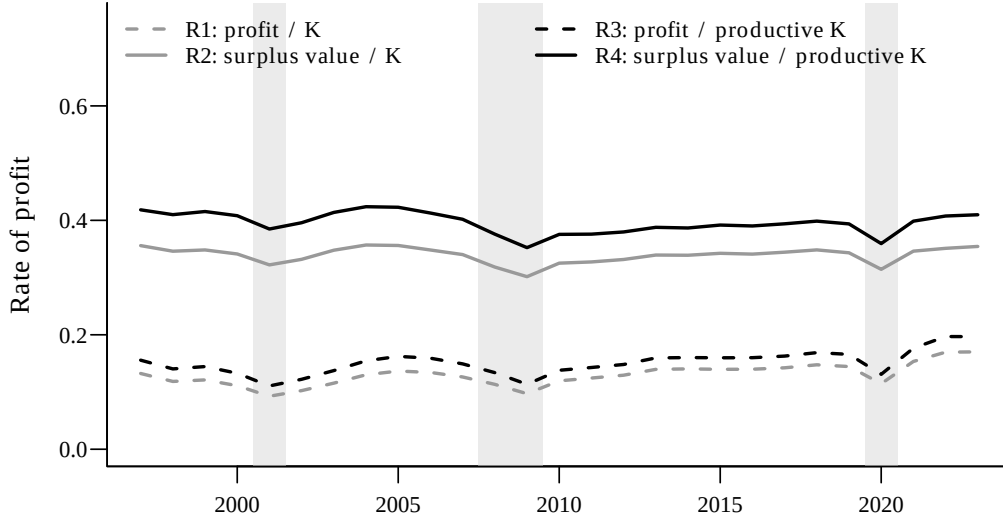


Figure 2: Four measures of the rate of profit, 1997–2023. Shaded areas are recession years.

Despite their different levels, the four measures move together, and all four fall in each of the three recessions in the sample (2001, 2008–2009, and 2020) and recover afterwards. The declines are proportionally larger for the measures based on profit than for those based on surplus value. In 2020, for example, surplus value fell by 6 percent whereas profit fell by 19 percent, because the compensation of unproductive labor, which is subtracted from surplus value to obtain profit, continued to rise. The same pattern appears in 2001, when surplus value fell by 1 percent and profit by 12 percent.

To check the level of R_1 , we compare it with a conventional rate of profit: the net operating surplus of private business (excluding housing and government), divided by the same capital stock. Figure 3 shows that R_1 , at 9–17 percent, lies below the conventional rate, at 13–16 percent, in every year until 2021 and slightly above it in 2022–2023. R_1 is lower because profit $\Pi = MNP - CE$ deducts the wages of all unproductive workers, including those in general government and in non-production industries, which are paid out of the Marxian net product. The two rates also behave differently over the cycle. In 2020 the conventional rate was unchanged at 15 percent, whereas R_1 fell from 14.5 to 11.5 percent, because the net product fell while the wages of unproductive labor continued to rise.

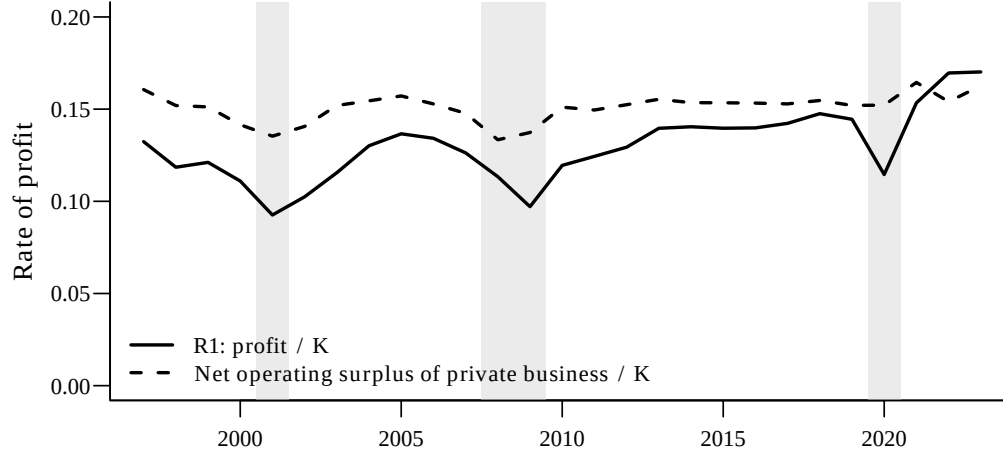


Figure 3: R_1 and the net operating surplus of private business relative to the capital stock, 1997–2023.

5.3 The Marxian net product and GDP

The Marxian net product is between 66 and 74 percent of GDP over 1997–2023. Table 5 decomposes the difference exactly. Relative to GDP, the Marxian net product excludes the value added of general government (10–13 percent of GDP), housing (9–10 percent), legal, management, and administrative services (5–6 percent), and finance net of the payments it receives from production (4–6 percent), as well as depreciation in production sectors (7–9 percent). It adds the intermediate inputs of the business sectors (9–14 percent of GDP), which the national accounts treat as costs of trade but which, in the Marxian accounts, are part of the value realized through trade.

Table 5: From GDP to the Marxian net product (percent of GDP)

	1997	2007	2009	2019	2020	2023
GDP	100.0	100.0	100.0	100.0	100.0	100.0
depreciation in production sectors	−7.3	−7.5	−8.1	−8.2	−8.6	−8.2
unproductive share of other services	−1.3	−1.1	−1.1	−1.1	−1.1	−1.1
intermediate inputs of business sectors	+9.1	+10.7	+8.6	+12.3	+11.8	+13.2
financial value added, net of payments from production	−4.8	−4.9	−4.5	−5.7	−6.1	−5.4
housing	−9.0	−9.3	−10.1	−9.2	−9.7	−9.5
legal, management, and administrative services	−4.9	−6.0	−5.7	−6.1	−6.0	−6.0
general government	−11.9	−11.9	−13.0	−11.2	−11.7	−10.4
Marxian net product	69.9	69.9	65.9	70.9	68.8	72.6
GDP (\$bn)	8,578	14,474	14,478	21,540	21,354	27,721

Notes: GDP is the sum of value added over all industries in the BEA Use tables at producers' prices. Each row is exact; the rows sum to the Marxian net product.

The two aggregates convey different information in three respects. First, the Marxian net product is considerably more volatile than GDP. The standard deviation of its annual growth rate is 4.3 percentage points, compared with 2.6 for GDP, and a regression of its growth on GDP growth gives

a slope of 1.5 (Figure 4). In 2009, nominal GDP fell by 2.0 percent while the Marxian net product fell by 6.5 percent; in 2020 the declines were 0.9 and 3.8 percent. The difference arises because the value added of general government, housing, and administrative activities, which GDP counts as output but which are financed out of value produced elsewhere, continued to grow in both recessions (by 0.3 and 2.6 percent). GDP therefore understates the contraction of the value-producing sphere in a crisis, which is the object of interest for theories of crisis based on profitability.

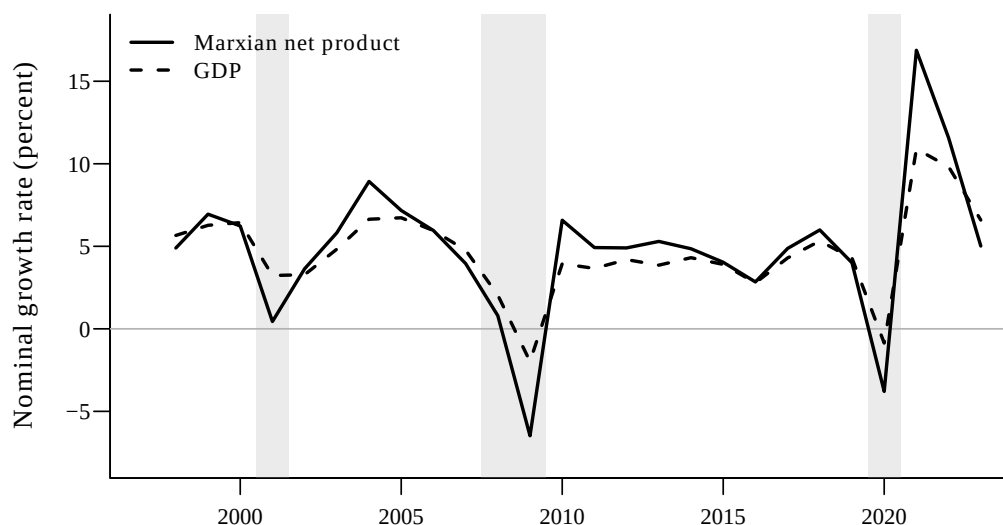


Figure 4: Nominal growth of the Marxian net product and GDP, 1998–2023. Shaded areas are recession years.

Second, the intermediate inputs of the business sectors rose from 9.1 percent of GDP in 1997 to 13.2 percent in 2023. A growing part of the social product is used up in realizing value rather than in producing it, a development that GDP, which nets out intermediate inputs, does not reveal.

Third, the Marxian accounts distinguish the rate of surplus value from the distribution of surplus value between profit and the wages of unproductive labor. As shown in Section 5.4, the share of surplus value absorbed by unproductive wages fell from about two-thirds to one-half, which explains why conventional profit rates rose over the period while the Marxian rate of profit did not. Conventional accounts, which treat all wages alike, cannot separate these two sources of movement in profitability.

5.4 Long-run movements, 1997–2023

Equation (4) allows us to attribute movements in the Marxian rate of profit to the rate of surplus value and the organic composition of capital. Figure 5 shows that both rise over the period. A regression of their logarithms on a linear trend, with Newey–West standard errors, gives an annual growth rate of 0.92 percent for the organic composition of capital ($t = 5.5$) and 0.72 percent for the rate of surplus value ($t = 4.5$). The organic composition of capital rises from 3.2 in 1997 to a peak

of 4.0 in 2011 and remains between 3.7 and 4.0 afterwards, while the rate of surplus value rises from 1.35 to 1.58. Because the two nearly offset each other, the Marxian rate of profit R_4 declines by only 0.20 percent a year, and the decline is not statistically significant at conventional levels ($t = -1.8$).

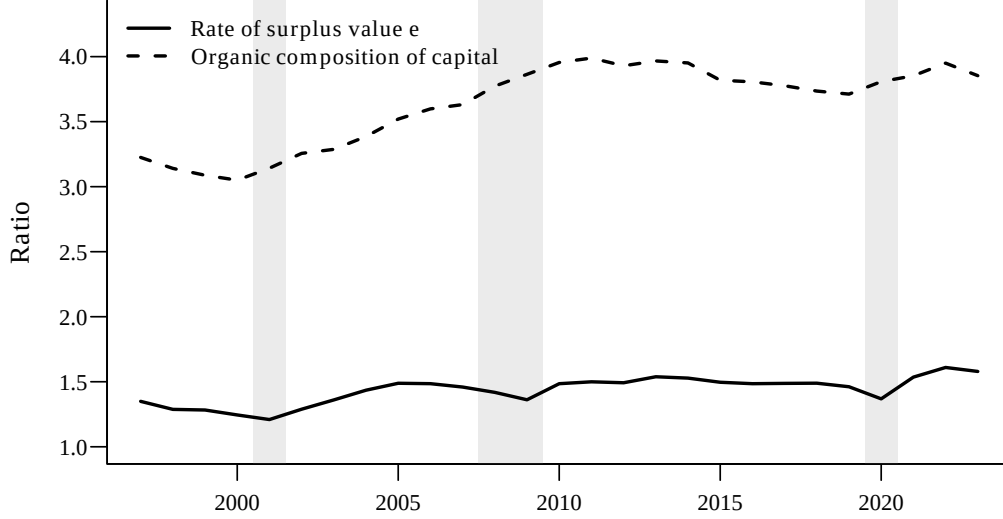


Figure 5: Rate of surplus value and organic composition of capital, 1997–2023. Shaded areas are recession years.

Table 6 applies the decomposition to subperiods. Over the whole period, the rise in the organic composition of capital lowered R_4 by 17.8 log points and the rise in the rate of surplus value raised it by 15.7, leaving a net decline of 2.1 log points. The subperiods show that the two forces operate at different frequencies. The rise in the organic composition of capital is concentrated in the expansion of 2001–2007 and the Great Recession, when it lowered R_4 by 14.5 and 6.2 log points. Movements in the rate of surplus value account for about half of the falls in R_4 in 2001 and 2008–2009, for most of the fall in 2020, and for all of the recovery after 2020.

Table 6: Decomposition of changes in the Marxian rate of profit (log points $\times 100$)

Period	$\Delta \ln R_4$	$\Delta \ln e$	$-\Delta \ln OCC$
1997–2000	−2.5	−8.1	+5.6
2000–2001	−5.9	−2.9	−3.0
2001–2007	+4.3	+18.8	−14.5
2007–2009	−13.2	−6.9	−6.2
2009–2019	+11.1	+7.1	+4.0
2019–2020	−9.1	−6.6	−2.6
2020–2023	+13.1	+14.3	−1.2
1997–2023	−2.1	+15.7	−17.8

Notes: $\Delta \ln R_4 = \Delta \ln e - \Delta \ln OCC$ holds exactly (equation 4).

These movements are consistent with the law of the tendency of the rate of profit to fall in its weak form: the organic composition of capital rises, and a rising rate of surplus value offsets most

of its effect on profitability. Over a period of 27 years, however, the resulting decline in the Marxian rate of profit is small relative to its cyclical fluctuations.

Figure 6 shows a second long-run change. The share of surplus value absorbed by the wages of unproductive labor fell from 0.63 in 1997 and 0.71 in 2001 to 0.52 in 2023, a decline of 0.8 percent a year ($t = -4.9$). Only about half of surplus value therefore remains as profit in the conventional sense, available for accumulation, interest, rent, and taxes, but that half has grown. Profit therefore rose faster than surplus value, and R_1 rose by 1.25 percent a year ($t = 3.6$) even as R_4 was stable or falling. Measures of profitability based on conventional profit can thus point in a different direction from the Marxian rate of profit over long periods, and not only in their levels. This is the mechanism emphasized by Paitaridis and Tsoulfidis (2012) and Tsoulfidis and Paitaridis (2019), operating here in reverse: the share of surplus value consumed by unproductive activities fell, enlarging the part available for accumulation.

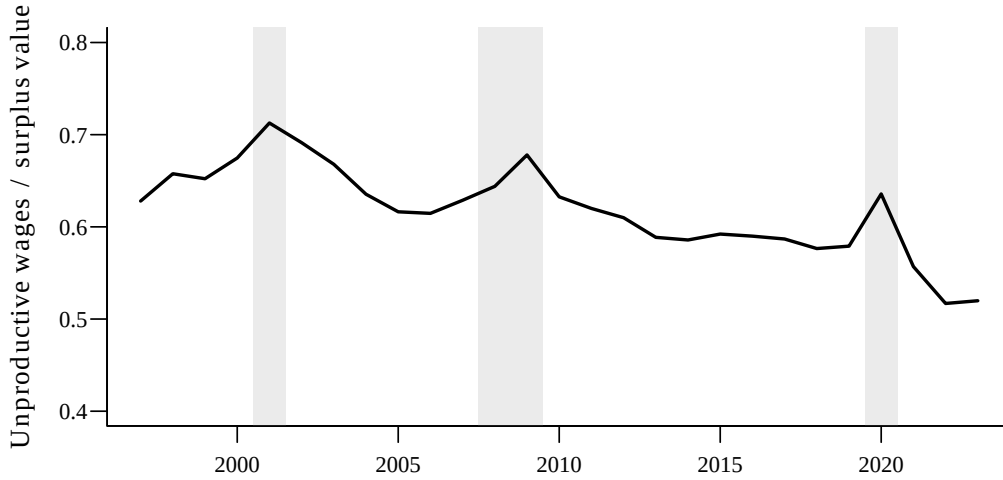


Figure 6: Share of surplus value absorbed by the wages of unproductive labor, 1997–2023. Shaded areas are recession years.

5.5 Sensitivity to measurement choices

Table 7 and Figure 7 report the long-run results under alternative measurement choices. Varying the productive share of other services between $\theta = 0$ and $\theta = 1$ changes the Marxian net product by 2.6–3.9 percent. Together with tracing payments to finance through the Make table, it changes the mean rate of surplus value by at most 0.03 and its trend by at most 0.03 percentage points a year. Measuring productive capital from depreciation shares rather than net stocks raises the level of R_4 by 16 percentage points and removes its small downward trend, but the annual changes of the two measures are almost perfectly correlated.

The classification of borderline industries matters more for the level of the rate of surplus value. Treating computer systems design and other professional, scientific, and technical services as non-production, as supporting services to business rather than production, raises the rate of

surplus value to 1.34–1.89. Treating educational services, health care, and social assistance as non-production, as activities of social reproduction rather than commodity production, raises it to 1.44–2.07. Because these industries are labor intensive, excluding them from production raises the rate of surplus value and its trend, to 1.0–1.1 percent a year. The Marxian rate of profit changes by at most 3 percentage points, because these reclassifications lower productive capital and productive compensation in similar proportions. In every specification the rate of surplus value rises, the organic composition of capital rises, and the Marxian rate of profit falls in each recession.

Table 7: Robustness of the long-run results, 1997–2023

Specification	e		R_4		Corr. of annual changes in R_4
	Mean	Trend (%/yr)	Mean	Trend (%/yr)	
Baseline	1.43	+0.72	39.6%	−0.20	1.000
Productive capital from depreciation shares	1.43	+0.72	55.5%	+0.02	0.985
same, residential depreciation excluded	1.43	+0.72	44.5%	+0.02	0.996
Financial payments by industry (Make table)	1.44	+0.71	39.7%	−0.22	1.000
$\theta = 0$	1.46	+0.75	39.9%	−0.18	1.000
$\theta = 1$	1.41	+0.69	39.2%	−0.23	1.000
Professional services as non-production	1.62	+1.01	39.6%	−0.17	0.998
Education and health as non-production	1.80	+1.09	42.7%	−0.13	0.997

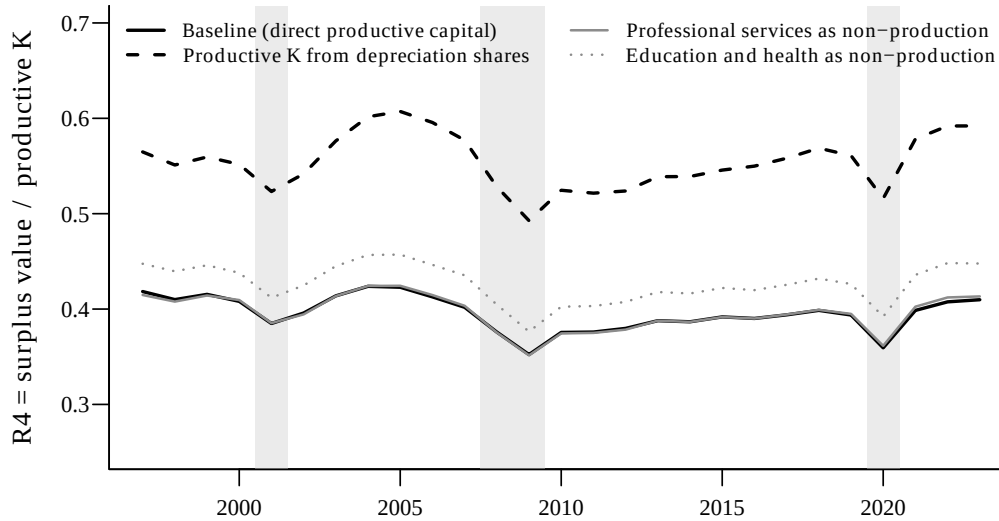


Figure 7: The Marxian rate of profit under alternative measurement choices, 1997–2023.

5.6 Comparison with previous estimates

Our estimates can be compared with those of Tsoulfidis and Paitaridis (2019), who apply a closely related framework to the NAICS-based U.S. accounts for 1964–2016. Their classification of industries is similar to ours. They also treat trade, real estate, finance, legal services, the management of

companies, administrative and support services, and general government as non-production, and education, health care, the arts, and accommodation and food services as production. Their estimated rate of surplus value, however, is much higher than ours, averaging 2.87 over 1964–2016, 3.65 over the last fifteen years of their sample, and 3.71 in 2016, compared with 1.21–1.61 over 1997–2023 in our estimates.

The difference arises mainly from the definition of variable capital. Tsoulfidis and Paitaridis (2019) count as variable capital only the wages of production workers within production industries, treating the wages of supervisory, managerial, and other overhead personnel in these industries as part of surplus value. We count the full compensation of employees in production industries as variable capital, in line with the industry-level approach of Shaikh and Tonak (1994), because the IO accounts do not divide compensation within industries by occupation. Our variable capital is therefore larger and our rate of surplus value correspondingly lower. The two approaches bracket the plausible range. Ours treats all labor in production industries as productive, while theirs treats only the direct labor of production workers as productive. Combining our industry classification with occupational data on production and nonsupervisory employment within industries would allow the two approaches to be reconciled.

6 Conclusion

This paper reconstructs Marxian categories from U.S. national and input-output accounts, following the framework of Shaikh and Tonak (1994) and applying it at the level of individual IO industries. We classify every industry into production, semi-production, business, financial, and non-production sectors, state the reasoning behind the classification of contested industries, identify the part of the net product that the official accounts record outside production, and construct the Marxian net product, surplus value, profit, the rate of surplus value, the organic composition of capital, and four measures of the rate of profit for 1997–2023.

The results show that the mapping from theoretical categories to data is quantitatively consequential. About two-fifths of the Marxian net product is recorded in business sectors rather than in production, a share that has grown since the late 1990s, and nearly half of all compensation is paid to unproductive labor. The Marxian net product is about 70 percent of GDP but considerably more volatile, so that GDP understates the contraction of the value-producing sphere in recessions. The level of the rate of profit ranges from about 9 to 42 percent depending on the treatment of unproductive labor and capital. Over 1997–2023, the organic composition of capital and the rate of surplus value both rose, the former slightly faster, so that the Marxian rate of profit shows a weak and statistically insignificant downward tendency and falls sharply in each recession. The share of surplus value absorbed by unproductive labor declined, so conventional profit rates rose even as the Marxian rate of profit did not.

These findings carry a methodological lesson for empirical work on the tendency of the rate of profit to fall. Conclusions about the level of profitability can depend on accounting choices as

much as on economic dynamics, so the correspondence between empirical variables and theoretical concepts deserves as much attention as econometric identification. Making the classification of each industry explicit, as we do here, allows these choices to be examined and their consequences to be measured.

The main limitation of the present analysis is that our variable capital includes the full compensation of employees in production industries. Combining our industry classification with occupational data would separate productive and unproductive labor within industries and reconcile our estimates with those of Tsoulfidis and Paitaridis (2019). A second extension is to relate the measured categories to accumulation and crisis more directly, using the longer series constructed here.

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A Classification of Industries

Table 8: Classification of BEA input-output industries

Production (52 industries)

Farms; Forestry, fishing, and related activities; Oil and gas extraction; Mining, except oil and gas; Support activities for mining; Utilities; Construction.

Wood products; Nonmetallic mineral products; Primary metals; Fabricated metal products; Machinery; Computer and electronic products; Electrical equipment, appliances, and components; Motor vehicles, bodies and trailers, and parts; Other transportation equipment; Furniture and related products; Miscellaneous manufacturing.

Food and beverage and tobacco products; Textile mills and textile product mills; Apparel and leather and allied products; Paper products; Printing and related support activities; Petroleum and coal products; Chemical products; Plastics and rubber products.

Air transportation; Rail transportation; Water transportation; Truck transportation; Transit and ground passenger transportation; Pipeline transportation; Other transportation and support activities; Warehousing and storage.

Publishing industries, except internet (includes software); Motion picture and sound recording industries; Broadcasting and telecommunications; Data processing, internet publishing, and other information services; Computer systems design and related services; Miscellaneous professional, scientific, and technical services.

Waste management and remediation services; Educational services; Ambulatory health care services; Hospitals; Nursing and residential care facilities; Social assistance; Performing arts, spectator sports, museums, and related activities; Amusements, gambling, and recreation industries; Accommodation; Food services and drinking places; Federal government enterprises; State and local government enterprises.

Semi-production (1 industry)

Other services, except government.

Business (7 industries)

Wholesale trade; Motor vehicle and parts dealers; Food and beverage stores; General merchandise stores; Other retail; Other real estate; Rental and leasing services and lessors of intangible assets.

Financial (4 industries)

Federal Reserve banks, credit intermediation, and related activities; Securities, commodity contracts, and investments; Insurance carriers and related activities; Funds, trusts, and other financial vehicles.

Non-production (9 industries)

Housing; Legal services; Management of companies and enterprises; Administrative and support services; Federal general government (defense); Federal general government (nondefense); State and local general government; Scrap, used and secondhand goods; Noncomparable imports and rest-of-the-world adjustment.
